Product Requirements Document (PRD)

Product Name

ExVeg

A multi-vendor agricultural marketplace where verified sellers can sell seeds, fertilizers, tools, and agricultural products to farmers and home gardeners.

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1. Product Overview

Problem Statement

Farmers and home gardeners face difficulties in purchasing agricultural products in small or customized quantities. Most platforms focus on bulk purchases and lack seller verification, making product quality unreliable.

Solution

Develop a secure multi-vendor e-commerce platform where:

- Sellers are verified before selling.
- Products are reviewed before publication.
- Buyers can purchase agricultural products safely.
- Home gardeners can buy customized combo packs and smaller quantities.
- Payments are handled securely through Razorpay.

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2. User Roles

Buyer

Can:

- Register/Login
- Browse Products
- Search Products
- Filter Products
- Add to Cart
- Checkout
- Track Orders
- Manage Profile
- View Order History
- Save Addresses

Seller

Can:

- Register/Login
- Submit Verification Documents
- Manage Profile
- Add Products
- Edit Products
- Manage Inventory
- View Orders
- Update Shipping Status
- View Earnings

Admin

Can:

- Approve/Reject Sellers
- Approve/Reject Products
- Manage Users
- Manage Categories
- Manage Orders
- View Analytics
- Handle Complaints
- Monitor Revenue

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3. Core Features

3.1 Seller Verification System

Flow

Seller Registration

↓

Pending Verification

↓

Admin Review

↓

Approved / Rejected

Seller Registration Fields

- Full Name
- Business Name
- Mobile Number
- Email
- Address
- GST Number (Optional)
- Aadhaar/PAN
- Bank Details
- Password

Admin Actions

- Approve Seller
- Reject Seller
- Request Additional Documents

Status

- Pending
- Approved
- Rejected

3.2 Product Verification System

Flow

Seller Upload Product

↓

Pending Product Review

↓

Admin Verification

↓

Approved / Rejected

Product Information

- Product Name
- Category
- Subcategory
- Description
- Price
- Discount
- Stock Quantity
- Images
- Weight
- SKU
- Brand
- Shipping Details

Special Field

Target Customer

Checkbox:

☐ Farmer

☐ Home Gardener

☐ Both

This field is important for filtering.

3.3 Home Gardener Special Feature

This is your USP (Unique Selling Point).

Example

Home gardener wants:

- Tomato Seeds (100 gm)
- Compost (2 kg)
- Small Gardening Tool Set

Seller can combine these products and ship them together.

Features

- Home Gardener Category
- Small Quantity Purchase
- Combo Pack Builder
- Customized Packaging
- Quantity Selector

3.4 Product Catalog

Categories

- Seeds
- Fertilizers
- Pesticides
- Gardening Tools
- Irrigation Equipment
- Organic Products
- Plant Growth Promoters
- Nursery Plants

Filters

- Category
- Price
- Rating
- Location
- Home Gardener
- Farmer
- Brand

3.5 Shopping Cart

Features

- Add to Cart
- Remove Product
- Update Quantity

- Save for Later
- Coupon Support

3.6 Checkout

Steps

Cart

↓

Address

↓

Payment

↓

Order Confirmation

Payment Gateway

Razorpay

Support:

- UPI
- Credit Card
- Debit Card
- Net Banking
- Wallets

3.7 Order Management

Buyer

- Order History
- Order Tracking
- Invoice Download
- Cancel Order
- Return Request

Seller

- View Orders
- Accept Order
- Process Order
- Generate Shipping Label
- Mark Shipped
- Mark Delivered

3.8 Inventory Management

Seller can:

- Add Stock
- Update Stock
- Receive Low Stock Alerts

3.9 Reviews and Ratings

Buyer can:

- Rate Product
- Review Product
- Upload Product Images

3.10 Notifications

Email

- Registration Success
- Seller Approval
- Product Approval
- Order Confirmation
- Shipment Updates

SMS (Future)

- OTP Verification
- Order Updates

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4. Screens

Public Screens

1. Home Page
2. About Us
3. Services
4. Products Listing
5. Product Details
6. Contact Us
7. Login
8. Registration

Buyer Screens

1. Dashboard
2. My Orders
3. Wishlist
4. Cart
5. Checkout
6. Address Management
7. Settings

Seller Screens

1. Dashboard
2. Product Management
3. Add Product
4. Inventory
5. Orders
6. Earnings
7. Profile
8. Settings

Admin Screens

1. Dashboard

2. Seller Verification
3. Product Verification
4. User Management
5. Order Management
6. Category Management
7. Reports
8. Analytics
9. Settings

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5. Database Design (High Level)

Users

```
{
 _id,
 role,
 name,
 email,
 mobile,
 password,
 status
}
```

Sellers

```
{
 userId,
 businessName,
 documents,
 verificationStatus
}
```

Products

```
{
 sellerId,
 productName,
```

category,

description,

price,

stock,

images,

targetAudience,

approvalStatus

}

Orders

{

buyerId,

products,

amount,

paymentStatus,

orderStatus

}

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6. Technology Stack

Frontend

- React.js
- Redux Toolkit
- Tailwind CSS
- Axios

Backend

- Node.js
- Express.js

Database

- MongoDB Atlas

Authentication

- JWT
- Refresh Tokens
- Role Based Access Control

Storage

- Cloudinary (Product Images)

Payment

- Razorpay

Deployment

Frontend

Vercel

Backend

Render

Database

MongoDB Atlas

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7. Future Enhancements

Phase 2

- AI Product Recommendation
- Farmer Community
- Seller Analytics
- Subscription Plans

Phase 3

- Android App
- iOS App
- Multiple Languages
- Live Chat Support

MVP (Version 1.0)

For the first release, focus only on:

- ✅ Buyer Registration/Login
- ✅ Seller Registration/Login
- ✅ Admin Seller Approval
- ✅ Product Listing
- ✅ Product Approval Workflow
- ✅ Product Search & Filters
- ✅ Home Gardener Filter
- ✅ Cart & Checkout
- ✅ Razorpay Payment
- ✅ Order Tracking
- ✅ Seller Order Management

Do not build AI features, chat, coupons, subscriptions, or mobile apps in Version 1.0. Build the marketplace first, validate it with users, then move to advanced features. This approach will save months of development time and reduce complexity significantly.